

Human Activity Recognition Using Feature Engineering + Machine Learning

Week-1 Deliverable Report

Submitted By:

Saima Ghulam Mustafa BSEF23M536

Uswah Zulfiqar BSEF23M548

1. Dataset Summary

Dataset Name

UCI HAR Dataset (Human Activity Recognition Using Smartphones)

Source

UCI Machine Learning Repository

Description

The UCI HAR dataset consists of smartphone motion sensor data. Each person performed **six daily activities**:

1. Walking
2. Walking Upstairs
3. Walking Downstairs
4. Sitting
5. Standing
6. Laying

Smartphone sensors used:

- **Accelerometer**
- **Gyroscope**

Signals were sampled at **50 Hz** in **128-length windows**, generating **561 features**.

Dataset Statistics

Property	Value
Total samples (train)	7,352
Total samples (test)	2,947
Number of features	561
Number of classes	6
Problem type	Multiclass Classification

2. Feature Engineering

Custom features were extracted directly from each 561-length sensor vector.

A. Statistical Features

- Mean
- Median
- Standard Deviation
- Variance
- Minimum
- Maximum
- Skewness
- Kurtosis

B. Temporal Features

- Zero Crossing Rate (ZCR)
- Signal Energy
- Peak-to-Peak (PTP) Amplitude
- Signal Magnitude Area (SMA)

C. Frequency Features

Computed using FFT:

- Spectral Centroid
- Spectral Entropy

3. Machine Learning Models

The following models were trained:

1. **Logistic Regression**
2. **Support Vector Machine (RBF Kernel)**
3. **Random Forest Classifier**
4. **XGBoost Classifier**

Data was split using an 80/20 train-test split and scaled using **StandardScaler**.

4. Evaluation Metrics

We evaluated models using:

- Accuracy
- Precision (weighted)
- Recall (weighted)
- F1-Score
- Confusion Matrix

5. Results

Performance Summary

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.6159	0.6123	0.6159	0.6060
SVM (RBF)	0.6084	0.6075	0.6084	0.5967
Random Forest	0.6390	0.6380	0.6390	0.6357
XGBoost	0.6485	0.6468	0.6485	0.6463

Best Model:

XGBoost achieved the highest accuracy: 64.85%

6. Confusion Matrix Interpretation

Strengths

- All models classify **walking-related activities** well.
- These activities have strong motion patterns → easier for ML to recognize.

Weaknesses

- Heavy confusion among:
 - **Sitting** ↔ **Standing**
 - **Standing** ↔ **Laying**
- These three activities produce very similar low-variance sensor readings.

7. Analysis & Discussion

Feature Importance Observations

- **Variance, Standard Deviation, Energy, SMA** strongly distinguished dynamic vs static activities.
- **Spectral entropy** improved class separation for rhythmic movements.
- **ZCR** helped differentiate walking vs non-walking classes.

Model Comparison

- Logistic Regression and SVM performed moderately.
- Random Forest improved accuracy due to non-linear modeling.
- **XGBoost performed best** because:
 - It handles non-linear feature interactions.
 - It deals well with noise.
 - It boosts weak learners into stronger ensembles.

Limitations

- Only ~13 engineered features contain significantly less information than raw time-series.
- Without full sequence data, the models miss temporal behavior.
- Static activities remain extremely hard to classify with classical ML.

8. Conclusion

- Custom feature engineering
- Four ML models
- Complete performance evaluation
- Analysis of strengths and limitations

Best performing model: **XGBoost (64.85% accuracy)**.

